Problem Statement:

Live news broadcasts, public media, and real-time conversations remain largely inaccessible to Deaf and hard-of-hearing audiences who rely on Indian Sign Language (ISL), as traditional human sign interpretation is expensive, resource-intensive, and impossible to scale across live streams or everyday communication. While closed captions help, they do not provide an accessible native-language experience for those who primarily use sign language. Furthermore, existing automated text-to-sign attempts often suffer from poor translation accuracy, forced unreviewed signs, and robotic or unnatural animations. There is a critical need for an automated, mobile-first solution that combines on-device sign recognition, conversation support, and a streaming live media translator delivering fluent, reviewed sign language sequences through a 3D avatar paired with synchronized captions.